

ParaEMT EMT-validation campaign (IAS2026 / TX4)

V2 report

Branch `research/parent-emt-validation` (not pushed)

2026-09-11

Status: V2 STOPPED at gate G3a.

- **V1 (unchanged, permanent):** G3 FAIL, G4 FAIL, EMT04–EMT18 BLOCKED.
- **V2** (preregistered at 0ab8efd4, before any V2 computation):
 - SW: PASS.
 - **G3a: FAIL.** The preregistered ringdown sub-check failed on the blind case. Every state trajectory agreed to $\leq 2.1 \times 10^{-4}$ of its excursion.
 - G3b, G4a, G4b: **NOT RUN** (stop rule). The GFL blind holdouts remain blind.
 - EMT04–EMT18: BLOCKED.
- **New finding.** The v1-frozen modal estimator A (multi-channel matrix pencil) is numerically unfit. On synthetic IEEE-39-type signals it crashes in 3 of 4 cases and is unresolved in the fourth.

Contents

1	Required wording	1
2	Side by side	2
3	V2-0 ledger reconciliation (commit cf8a0b38)	2
4	V2 design	2
5	Gate SW: PASS	3
6	Gate G3a: FAIL	3
7	Diagnosis (post hoc)	3
8	What was not run, and why	4
9	Claim ledger	4
10	For the author: what a V3 would need	4
11	Reproducibility	5
	What EMT did and did not validate	5

1 Required wording

The first preregistered EMT realization failed its device-interface gates. A second preregistered realization, designed after diagnosing those failures and evaluated on new holdouts, stopped at its synchronous-machine transcription gate. Every machine state trajectory agreed with the frozen phasor model to 2×10^{-4} of its excursion. The preregistered ringdown sub-check, which reused the frozen v1 matrix-pencil estimator, returned a spurious mode for both the EMT run and its phasor reference. The grid-following-converter gates were not reached, and no portfolio result was produced.

The V1 FAIL is not replaced, and there is no V2 PASS.

2 Side by side

element	V1 outcome	post hoc V1 diagnosis	V2 prospective design	V2 blind-holdout result	final EMT result
network / operating point	G1 PASS (Ybus 1.7×10^{-16} ; V 1.5×10^{-6})	—	unchanged	—	reproduced (V19/I01)
SG transcription	not separated (G3)	exact to 1.9×10^{-4} with an algebraic line (D1)	G3a: algebraic network; bus 30 regression + bus 36 blind; v1 rules	trajectories PASS (bus 36: $\leq 2.1 \times 10^{-4}$); ringdown sub-check FAIL (spurious estimator mode in both EMT and reference; $\Delta\alpha$ $0.016 > 0.005$)	G3a FAIL (preregistered rule)
SG electromagnetic embedding	G3 FAIL (e_{fd} $2.26\% > 2\%$)	the line's $(X/\omega_0)dI/dt$ (D3: 1.4×10^{-3} against a dynamic-line reference)	G3b: against a dynamic-line reference	NOT RUN	not established
GFL transcription	not separated (G4)	exact to 0.78 % with an algebraic line (D1)	G4a: v1 harness; bus 35 and 37 blind	NOT RUN (holdouts blind)	not established
GFL EMT interface	G4 FAIL (ideal current source; Nyquist growth)	current source into a lossless trapezoidal inductor (D4)	G4b: voltage source behind R_f-L_f ; 6 checks	NOT RUN (holdouts blind); software gate SW PASS	not established
modal estimator (frozen v1 §5)	never exercised on IEEE-39	—	reused unchanged	unresolved on every unit ringdown; crashes on 3/4 synthetic multi-channel signals (DV2-2)	unfit as frozen
EMT04–EMT18 portfolio	BLOCKED	—	resume unchanged if all gates pass	—	BLOCKED

3 V2-0 ledger reconciliation (commit cf8a0b38)

- **I04/V21.** The final ledger records the custom-GFL results as VALIDATED: reproduced in ANDES with the same equations (post-cumulant Phase 8, GATE 5 PASS). The v1 EMT report had wrongly said “NOT YET TESTED”. Independent phasor reproduction and EMT reproduction are now kept apart. EMT reproduction was not achieved, in v1 or in V2.
- **T08.** The Kundur holdout scores 28/29 under the preregistered rule. The miss is a real-pair coalescence, so the score is 28/28 for origin crossings.
- **Claim matrix.** It was regenerated from the final canonical ledger: 43 rows, none with EMT support.

4 V2 design

Documents (commit 0ab8efd4): docs/20260911_PAREMT_EMT_PREREG_V2.md and docs/20260911_PAREMT_GFL_VOLTAGE_INTERFACE_DERIVATION.md.

Gates. SW $\rightarrow$ G3a $\rightarrow$ G3b $\rightarrow$ G4a $\rightarrow$ G4b. Any failure stops the sequence.

GFL interface.

- An average-value voltage source behind the physical R_f-L_f filter.
- The measured filter currents are i_d and i_q ; the controller integrates 9 states.
- Command: $e_{\text{cmd}} = e_{\text{TX4}} + j(x_f/\omega_0)\theta'i$, evaluated at the predicted states and at linearly extrapolated phasors.
- No numerical damping.

Blind holdouts, frozen before any run:

- SG: bus 36 at its canonical dispatch;
- GFL: bus 35 at $S = 2 + j0.5$;
- GFL: bus 37 at its canonical dispatch.

Code. Commit f8b8fb15. The v1 kernel is untouched.

5 Gate SW: PASS

test	criterion	result
T1: (P) at e_{cmd} equals $f_{i,\text{TX4}}$ (2000 random vectors, $\theta' \in [-5, 5]$ rad/s)	$\leq 10^{-12}$ rel.	9.1×10^{-17}
T1: $e_{\text{cmd}} = e_{\text{TX4}} + j(x_f/\omega_0)\theta'i$	$\leq 10^{-12}$	2.2×10^{-16}
T2: abc-level physical filter vs dq ODE (0.2 s)	$\leq 10^{-7}$	2.2×10^{-12} (excursion 6.3×10^{-3})
T3: companion vs ParaEMT <code>numba_InitNet</code>	$\leq 10^{-12}$	1.8×10^{-16}
T4: static bases and signs at $t = 0$	$\leq 10^{-12}$	$\leq 3.0 \times 10^{-16}$

6 Gate G3a: FAIL

Run. Algebraic network, the v1 SG kernel, 50 μs , $P_m \times 1.02$ on $[1.0, 1.2)$ s, 10 s. The reference is the quasi-static phasor model.

quantity	rule	R-SG30 (regression)	B-SG36 (blind)
equilibrium $\Delta P, \Delta Q$ at 0.9 s	$\leq 10^{-4}$	0, 0	0, 0
worst state error/excursion (7 states)	≤ 0.02	1.9×10^{-4} (ω)	2.1×10^{-4} (ω)
frozen <code>matrix_pencil</code> on $\omega - 1$: $\alpha_{\text{EMT}}/\alpha_{\text{ref}}$	$ \Delta\alpha \leq 0.005$	-0.118368/ - 0.118367	+6.3134/ +6.2976 ($\Delta\alpha$ 0.0158)
$f_{\text{EMT}}/f_{\text{ref}}$	$ \Delta f \leq 0.005$	1.177112/1.177112	1.1051/1.1097 (Δf 0.0046)
case verdict	all rules	PASS	FAIL

Gate G3a: FAIL. The stop rule applies: G3b, G4a and G4b were not run, and EMT04–EMT18 stay BLOCKED. No threshold was changed and nothing was retuned.

7 Diagnosis (post hoc)

DV2-1: the physical ringdown agrees; the failure is in the estimator.

- The frozen `matrix_pencil` reports **resolved = False** in all four evaluations: both cases, EMT and reference alike.
 - Its reconstruction residual is 0.9987–0.9989, with model order 10 or 16.
 - On B-SG36 it selects a spurious growing mode, $\alpha \approx +6.3 \text{ s}^{-1}$, for the reference and the EMT run alike. That mode is not physical: the trajectories decay.
- An independent fixed-order damped-sinusoid fit of $\omega - 1$ over $[1.2, 10]$ s gives:

case	α_{EMT}	α_{ref}	$f_{\text{EMT}} = f_{\text{ref}}$
R-SG30	-0.1182316	-0.1182316	1.177050 Hz
B-SG36	-0.2439564	-0.2439563	1.250783 Hz

EMT and the reference agree on the physical mode to about 10^{-7} s^{-1} . The sub-check failed because the frozen estimator is unresolved, which the preregistered G3a rule did not require it to be. It did not fail because of a model discrepancy.

- v1.** The v1 EMT02 ringdown “agreement” came from the same unresolved estimator; its value there happened to be the physical one.
- V2.** “Where resolved” logic was preregistered only for the G4b estimator R. G3a keeps the unconditional v1 rule and is judged by it.

DV2-2: the frozen estimator on synthetic multi-channel data. The input imitates EMT05–EMT18: 20 channels, 30 s, a target mode plus a damped 0.95-Hz mode, and the pulse at 1 s.

true α, f	estimator A (matrix pencil)	estimator B (Hilbert)	classify
+0.127, 0.622 Hz	crash (LinAlgError; Vandermonde overflow)	resolved, +0.1266, 0.6216 Hz	ERROR
−0.144, 0.637 Hz	unresolved (−0.1440, 0.6370 Hz)	resolved, −0.1468	UNRESOLVED
−0.0174, 0.71 Hz	crash	unresolved, −0.0190	ERROR
+0.0036, 0.705 Hz	crash	unresolved, +0.0021	ERROR

Mechanism. The model order comes from the singular-value threshold 10^{-4} , and it admits spurious modes with $|z| \gg 1$. **z**k** then overflows in the least-squares residue fit, which gives inf, then NaN, then an SVD failure. When it does not crash, the spurious columns drive the residual to about 1, so the case is never “resolved”.

Consequence. Estimator A was frozen in v1 §5 and reused unchanged in V2, and its α_A is the reported α_{EMT} . It could not have produced a STABLE/UNSTABLE verdict for EMT05–EMT18 even if every device gate had passed. This is a defect of the frozen v1 design, found only now, and under the rules it is not repaired inside V2.

8 What was not run, and why

G3b, G4a and G4b were not run (V2 prereg §5.1, stop rule).

- The GFL blind holdouts (bus 35, $S = 2 + j0.5$; bus 37, canonical dispatch) have never been simulated in EMT.
- Only their phasor references, computed as frozen predictions, exist. Both GFL holdouts therefore stay prospective.
- The voltage-source interface was verified only at the software level (gate SW). It has not been run in any network.

EMT04–EMT18 remain BLOCKED, and no IEEE-39 EMT portfolio run exists.

9 Claim ledger

There is no change: **results/20260911_EMT_CLAIM_MATRIX.csv** (V2-0) stands.

- No claim receives EMT support, and none is EMT-refuted.
- V19 remains the only EMT QUANTITATIVELY REPRODUCED claim.
- V20 and V21 keep their final-ledger status. That status rests on the independent ANDES phasor reproductions.

10 For the author: what a V3 would need

This is a decision memo, not executed.

1. **Replace or repair estimator A before any use.** Freeze a numerically safe pencil and pre-validate it on known signals, then commit the validation before any TX4 EMT result.
 - *Validation set:* synthetic multi-channel signals (including DV2-2), and the frozen phasor TDS of P4 H4 and G_S , where it must reproduce the frozen eigenvalues.
 - *Example safeguards:* scaled residue evaluation, a model order fixed by an information criterion, and spurious-mode rejection.
2. **SG ringdown sub-check.** Condition it on the estimator being resolved, or use a preregistered fixed-order fit. Keep the 2% trajectory rule unchanged.
3. **Blind SG holdout.** B-SG36 has now been seen, so a V3 needs a new one. The GFL holdouts B1-GFL35 and B2-GFL37 are still blind and can be kept.
4. **Everything else unchanged:** the V2 interface, derivation and G4b checks; EMT04–EMT18, their frozen predictions and their success rules.
5. **Stop rule unchanged.** If the voltage interface fails G4b on the blind holdouts, STOP. There is no other interface in the same campaign.

11 Reproducibility

Rerun from commit f8b8fb15:

1. EMT_setup_tx4.py;
2. v2/EMTV2_refs.py (tx3-analysis);
3. v2/EMTV2_sw_tests.py;
4. v2/EMTV2_gates.py;
5. v2/EMTV2_G3a_diagnostics.py.

Byte identity. The chain was executed twice, before the code commit and again from the committed HEAD. The outputs are identical apart from provenance fields: see `results/EMTV2/EMTV2_rerun_identity.json`.

Provenance. `results/EMTV2/EMTV2_gates.json`, `SW/sw_tests.json` and `refs/refs_manifest.json` record the git HEAD, ParaEMT d79d735a, the working-copy diff SHA, the environment hash and the file sha256.

What EMT did and did not validate (after V1 and V2)

The table separates **theorem validity** from **model-fidelity corroboration**. Statuses are those of the final canonical ledger.

Theory / claim	Mathematical status	Phasor evidence	Independent phasor evidence	EMT evidence (V1 + V2)	Final status
V01, V02, V08, V22, V26, T04, T06, T08 (theorems)	proved	yes	where applicable	not applicable	PROVED
V03 P4 trap, $\kappa = 4$ [B02]	—	PCV02; G2 TDS	ANDES Phase 8, 32/32	BLOCKED	VALIDATED (model-specific witness)
V06, V07 policy dependence [B01]	structure proved	clean g -only counterfactual	ANDES Phase 8, P4 vs G_S	BLOCKED	VALIDATED
V10, V11 port derivative, line ranking	proved (V10)	PCV04	ANDES R3 11/12, ρ 1.00	BLOCKED	PROVED + VALIDATED; VALIDATED (local)
V13, V14 remediation, governors	—	PCV04; FC03; G2	partial ANDES	BLOCKED	VALIDATED (model-specific); BENCHMARK-SPECIFIC
V15–V18 robustness	—	PCV05	—	BLOCKED / not targeted	as in the ledger
V30, B07, B10 nonlinear scope	normal form	G2 TDS 32/32	—	BLOCKED	NUMERICALLY VALIDATED
V19 network and operating point [I01]	—	reference	ANDES; pandapower after the tap fix	EMT QUANTITATIVELY REPRODUCED (V1 G1)	CONDITIONAL (pandapower translation)
V20 SG dynamics [I02, I03]	—	internal	ANDES R2/R3, E1	V1 G3 FAIL; V2 G3a FAIL (trajectories 2×10^{-4} ; estimator sub-check) $\rightarrow$ EMT UNRESOLVED	VALIDATED (by ANDES)
V21 custom GFL in a second simulator [I04]	—	project DAE	ANDES Phase 8, GATE 5 PASS	V1 G4 FAIL; V2 G4a/G4b NOT RUN (blind) $\rightarrow$ EMT UNRESOLVED	VALIDATED as a reproduction of computation; EMT reproduction not achieved
cumulant and closure diagnostics, V05, V12, B04, B06, negatives	as in the ledger	as in the ledger	—	not applicable / BLOCKED	unchanged